

Project: House Price Prediction Using Supervised Learning

Objective:

Build a machine learning model to predict house prices and compare the performance of different regression algorithms.

Project Steps:

1. Data Preparation

- Load the California Housing dataset.
 - Analyze the dataset:
 - Number of rows and columns
 - Data types
 - Missing values
 - Outliers
 - Separate features (X) and target variable (y).
 - Split the data into:
 - 80% Training data
 - 20% Testing data
 - Apply feature scaling using StandardScaler.
-

2. Build Baseline Models

Implement and train the following models:

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor

For each model:

- Train the model using the training data.
 - Make predictions on the test data.
-

3. Model Evaluation

Evaluate each model using the following metrics:

- MAE (Mean Absolute Error)
- MSE (Mean Squared Error)
- RMSE (Root Mean Squared Error)
- R^2 Score

Create a Pandas DataFrame to compare the results.

Also, create bar charts to visualize and compare model performance.

4. Random Forest Hyperparameter Tuning

Optimize the Random Forest model using:

- GridSearchCV
- or
- RandomizedSearchCV

Tune at least two hyperparameters:

- `n_estimators`
- `max_depth`

Compare the optimized model with the original Random Forest model.

5. Feature Importance Analysis

- Extract feature importance values from the Random Forest model.
 - Plot a feature importance chart.
 - Identify which feature has the highest impact on house prices.
-

Final Deliverable:

Submit a Jupyter Notebook (`.ipynb`) containing:

- Clean and organized code
- Markdown explanations
- Evaluation charts
- Model comparison table
- Final conclusion and insights

